SEC P vs NP log 2026-04-23

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 04:32 UTC

SEC P vs NP — daily log 2026-04-23

9 cycles recorded

Homotopy Type of Random 3-SAT Clause Complexes and Satisfiability Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial homotopy theory × Random 3-SAT instances
- **Recorded**: 2026-04-23 21:40 UTC
- **Entry ID**: `5c256dcd7a85`

Statement

Let X(n, m) be the abstract simplicial complex where each clause in a 3-SAT formula on n variables is a 2-simplex, and faces correspond to variable literals. For random 3-SAT instances with $m = \lfloor cn \rfloor$ clauses, the complex has nontrivial fundamental group with high probability if and only if $c > 4.2$, and the instance is unsatisfiable. Moreover, if the fundamental group $\pi_1(X(n, m))$ is trivial, the formula is satisfiable with probability ≥ 0.95 for $n \geq 10$.

Rationale

The clause structure of 3-SAT can be interpreted as a 2-dimensional complex where topological connectivity may reflect logical consistency. Nontrivial π_1 could signal conflicting constraint loops that obstruct satisfiability. Prior work on clause complexes uses homology; homotopy is finer and sensitive to clause interactions beyond cycles.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

test_code_gen_failed

Kazhdan-Lusztig Polynomial Coefficients Bound Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Kazhdan-Lusztig theory of Hecke algebras × Resolution proof width
- **Recorded**: 2026-04-23 21:43 UTC
- **Entry ID**: `20680d0498ef`

Statement

For every unsatisfiable 3-CNF formula φ with n variables and m clauses, the maximum absolute coefficient of the Kazhdan-Lusztig polynomial associated to the Bruhat interval induced by its clause hypergraph is Ω(√w), where w is the minimal resolution width of φ.

Rationale

Kazhdan-Lusztig polynomials encode deep symmetry and singularity data in Coxeter systems, which may reflect logical 'obstruction thickness' in proof complexity. The Bruhat interval derived from clause-variable incidence offers a rare algebraic scaffold over SAT instances. This link could reveal how non-local logical dependencies manifest in representation-theoretic invariants.

Novelty

- Judge: `NOVEL` over 10 arXiv hits

Top hits consulted: - [0810.5165v2] Some computations about Kazhdan-Lusztig cells in affine Weyl groups of rank 2 - [1309.7265v2] Computing individual Kazhdan-Lusztig basis elements - [2007.15349v1] The inverse Kazhdan-Lusztig polynomial of a matroid - [1406.5202v3] Bruhat Interval Polytopes - [2307.15726v2] Subexpressions and the Bruhat order for double cosets

Empirical Test

- exit code: `0`, elapsed: `0.02s`

```
Conjecture: max |coeff| of KL poly is Ω(√w) for unsat 3-CNF φ
However, no canonical method to associate a Bruhat interval or Coxeter system to a 3-
CNF clause hypergraph.
Kazhdan-Lusztig polynomials are not defined for arbitrary hypergraphs or CNF formulas.
The mapping from 3-CNF to Bruhat interval is not specified in literature or constructively in the conje
Algebraic scaffold (Coxeter system) from clause-variable incidence is undefined.
RESULT: INCONCLUSIVE mapping from 3-CNF to Bruhat interval undefined
```

Judge reasoning

The mapping from 3-CNF to Bruhat interval is not specified, making it impossible to test the conjecture. | next: Define a canonical method to associate a Bruhat interval or Coxeter system to a 3-CNF clause hypergraph.

Betti Numbers of Independence Complexes Bound Resolution Width

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial algebraic topology × Resolution proof width
- **Recorded**: 2026-04-23 21:43 UTC
- **Entry ID**: `9c3bc7d2f5cf`

Statement

For every unsatisfiable 3-CNF formula F with n variables and m clauses, the total sum of reduced Betti numbers of the independence complex of its variable-clause incidence graph is at least the minimum width of any resolution refutation of F minus 1. This inequality is tight for minimally unsatisfiable formulas with connected incidence graphs.

Rationale

The independence complex captures higher-order dependencies among variable appearances, and its homology reflects combinatorial obstructions to satisfiability. Resolution width measures how globally the refutation must reason; high Betti numbers may signal many obstructions, forcing wide derivations. This provides a topological lower bound that is sensitive to clause-variable structure, not just counts.

Novelty

- Judge: `NOVEL` over 15 arXiv hits

Top hits consulted: - [1101.4831v1] The f–vector of the clique complex of chordal graphs and Betti numbers of edge ideals of uniform hypergraphs - [2111.02551v4] Bigraded Betti numbers and Generalized Persistence Diagrams - [0001101v4] Integrality of L2-Betti numbers - [2208.13438v3] Intermediate Ricci curvatures and Gromov's Betti number bound - [2501.12623v1] Betti number bounds for varieties and exponential sums

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

test_code_gen_failed

Jones Polynomial of Writhe-Weighted SAT-Link Diagrams Predicts Resolution Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Knot theory (quantum invariants) × Resolution proof size
- **Recorded**: 2026-04-23 21:47 UTC
- **Entry ID**: `192f4533ba68`

Statement

For any unsatisfiable 3-CNF formula φ with n ≤ 20 variables, map its variable-clause incidence structure to a signed link diagram D_φ via a clause-as-crossing coding. Let V_φ(t) be the Jones polynomial of D_φ evaluated at t = i. Then |V_φ(i)| = Θ(log R(φ)), where R(φ) is the minimal resolution proof size of φ.

Rationale

The Jones polynomial captures global topological entanglement, which may reflect the logical interdependency among clauses. Unsatisfiability requires deriving a contradiction through such entangled variable paths, potentially mirrored in knot crossings. The writhe weighting enforces orientation consistency with literal polarities.

Novelty

- Judge: `NOVEL` over 15 arXiv hits

Top hits consulted: - [9811006v1] Billiard knots in a cylinder - [0404143v1] Groups of locally-flat disk knots and non-locally-flat sphere knots - [0806.3452v2] Conjugate Generators of Knot and Link Groups - [9805078v3] Positive knots, closed braids and the Jones polynomial - [2510.12675v1] Centralizers of discrete Temperley-Lieb-Jones subfactors

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

test_code_gen_failed

Grothendieck-Witt Class of Clause Polynomials Mod 2 Predicts Resolution Width

- **Verdict**: `SUPPORTED`
- **Bridge**: Quadratic forms over finite fields (Grothendieck-Witt groups) × Resolution proof width
- **Recorded**: 2026-04-23 21:56 UTC
- **Entry ID**: `15ae8fd62af0`

Statement

For every 3-CNF formula φ with n variables and m clauses, compute the clause-indicator quadratic form Q_φ over 𝔽_2 defined by Q_φ(x) = Σ_{c∈φ} ⟨c, x⟩² mod 2. The 2-rank of the associated symmetric bilinear form B(x,y) = Q_φ(x+y) - Q_φ(x) - Q_φ(y) equals the minimum resolution width of φ up to an additive constant: |width_min(φ) - rank_2(B)| ≤ 3.

Rationale

The clause-indicator quadratic form captures parity interactions between variable assignments and clause violations. The Grothendieck-Witt class over 𝔽_2 encodes isotropy and metabolicity, which may reflect clause redundancy and variable pivoting in resolution. Low-rank forms may correspond to wide resolvents due to high degeneracy in conflict structure.

Novelty

- Judge: `NOVEL` over 14 arXiv hits

Top hits consulted: - [2201.00728v2] Grothendieck–Witt Groups of Henselian Valuation Rings - [1405.2700v1] Zero Excess and Minimal Length in Finite Coxeter Groups - [0610012v1] Tevatron-for-LHC Report of the QCD Working Group - [0710.5926v2] Mod 2 cohomology of 2-local finite groups of low rank - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation

Empirical Test

- exit code: `0`, elapsed: `0.76s`

```
Testing n=3...
Testing n=4...
Testing n=5...
Testing n=6...
RESULT: SUPPORTED max_diff=2
```

Judge reasoning

The test results show that the maximum difference between the minimum resolution width and the 2-rank of the bilinear form is 2, which is within the conjectured bound of 3. This provides strong evidence in favor of the conjecture. | next: Test larger instances ($n \geq 7$) to check if the difference remains bounded by 3 and investigate whether the bound is tight.

Clifford Algebra Rank of Boolean Functions Bounds Circuit Depth

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Clifford algebras over finite fields × Boolean circuit depth
- **Recorded**: 2026-04-23 21:59 UTC
- **Entry ID**: `6fc6466b2d76`

Statement

For any boolean function f on $n \leq 20$ variables given by its truth table, let C_f be the 2^n × 2^n matrix over $\mathbb{F}$_2 defined by (C_f)*{x,y} = f(x) · (-1)^{|x∧y|}. Then the rank of C_f over the Clifford algebra Cl*{n}($\mathbb{F}$_2) is Θ(depth(f)), where depth(f) is the minimum depth of a fan-in 2 circuit computing f.

Rationale

Clifford algebras encode geometric structure of boolean hypercubes via anticommutation relations, which may reflect information flow in shallow circuits. The matrix C_f combines Fourier-analytic and geometric data in a way sensitive to functional composition depth. This noncommutative rank could detect bottlenecks in circuit depth better than linear algebra over $\mathbb{F}$_2.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [0002194v2] On q-Deformations of Clifford Algebras - [0603050v1] Extended Grassmann and Clifford algebras - [1604.06967v1] Sparse Representations of Clifford and Tensor algebras in Maxima - [2011.05171v4] Subgroups of Clifford algebras - [1712.05384v2] Simulation of low-depth quantum circuits as complex undirected graphical models

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

test_code_gen_failed

Toric Degeneration Depth of SAT Polytopes Equals Decision Tree Complexity

- **Verdict**: `FALSIFIED`
- **Bridge**: Toric geometry (via Gröbner degenerations) × Boolean decision tree complexity
- **Recorded**: 2026-04-23 22:01 UTC
- **Entry ID**: `84f371b65a13`

Statement

For any Boolean function f represented as a CNF formula φ with n variables, let P_φ be the convex hull of its satisfying assignments. Let d(φ) be the minimum number of distinct Gröbner degenerations needed to transform P_φ into a toric variety with monomial ideal matching the clause structure. Then the decision tree complexity of f equals the minimum d(φ) over all term orders. In particular, D(f) = Θ(d(φ)).

Rationale

Toric degenerations capture how tightly the geometry of satisfying assignments is structured under monomial reductions. Decision tree complexity measures adaptive query efficiency, which may correspond to how many 'flat limits' are needed to expose clause-driven symmetries. The algebraic sparsification in toric degenerations mirrors variable fixing in decision trees.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `1.05s`

```
Testing n = 1
  1 formulas
Testing n = 2
  7 formulas
Testing n = 3
  63 formulas
Testing n = 4
  1023 formulas
  tested 100 formulas
  tested 200 formulas
  tested 300 formulas
  tested 400 formulas
  tested 500 formulas
  tested 600 formulas
  tested 700 formulas
  tested 800 formulas
Counterexample: n=4, clauses=((0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (3, 3)), D(f)=1, d(φ)=3
  truth_table has 4 satisfying assignments
RESULT: FALSIFIED counterexample_found
```

Judge reasoning

A counterexample was found for n=4 with D(f)=1 and d(φ)=3, showing that the decision tree complexity does not equal the toric degeneration depth. This violates the conjectured equivalence. | next: Analyze the structure of the counterexample formula to identify why d(φ) is larger than D(f); consider whether d(φ) can be bounded by a function of D(f) instead of being equal.

Tate-Shafarevich Group Order of Elliptic Curve Models SAT Instance Density

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Arithmetic geometry (Tate-Shafarevich groups) × 3-SAT instance phase transition
- **Recorded**: 2026-04-23 22:13 UTC
- **Entry ID**: `8d9b58ba4ae0`

Statement

For a random 3-SAT instance with n variables and m clauses, map it via a degree-2 rational model to an elliptic curve E over Q; then $|\text{Ш}(E)[3]| = \Theta((m/n - 4.26)^2 * n)$ for instances near the satisfiability threshold ($3.9 \le m/n \le 4.5$). A single counterexample with $n \le 20$ where the measured Tate-Shafarevich 3-torsion order deviates by more than 20% from this relation falsifies the conjecture.

Rationale

The phase transition in 3-SAT may reflect arithmetic instability in associated quadratic models. Elliptic curves with large Tate-Shafarevich groups exhibit hidden complexity analogous to unsatisfiable cores. The 3-descent obstruction mirrors local-global inconsistency in clause satisfaction.

Novelty

- Judge: `NOVEL` over 12 arXiv hits

Top hits consulted: - [1912.12928v1] Relating Tate-Shafarevich group of an elliptic curve with class group - [0303143v2] The p-part of Tate-Shafarevich groups of elliptic curves can be arbitrarily large - [1602.02932v1] Structure of Tate-Shafarevich groups of elliptic curves over global function fields - [2505.01628v1] The Satisfiability Threshold for K-XOR Games - [1307.7691v3] On the class numbers of the fields of the p^n-torsion points of certain elliptic curves over Q

Empirical Test

- exit code: `0`, elapsed: `0.02s`

```
Testing conjecture for n=10, m in range(39, 46)
However, no canonical mapping from 3-SAT to elliptic curve is known.
Constructing genus-one curves from 3-SAT via symmetric quadratics in P^3 is not a defined procedure.
Computing 3-Selmer group via 3-descent by isogeny in pure Python is infeasible without algebraic numbe
RESULT: INCONCLUSIVE mapping_and_computation_not_defined
```

Judge reasoning

The test was unable to map 3-SAT instances to elliptic curves or compute the 3-Selmer group due to lack of defined procedures and computational infeasibility. | next: Implement a canonical mapping from 3-SAT to elliptic curves and utilize algebraic number theory libraries for 3-descent computations.

Poincare Series

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Langlands correspondence × Communication complexity

- **Recorded**: 2026-04-23 22:57 UTC
- **Entry ID**: `121736aabe59`

Statement

For any boolean function f, the Poincare series of the Langlands dual of the Galois representation associated to f's communication matrix has a pole at s=1 if and only if f has a monochromatic rectangle. The order of the pole is bounded by the logarithm of the size of the smallest such rectangle.

Rationale

The Langlands correspondence provides a bridge between number theory and representation theory, and its geometric version can be used to study the representation theory of algebraic groups. This can be applied to the study of boolean functions by associating a Galois representation to the communication matrix of the function. The Poincare series of the Langlands dual of this representation may capture information about the combinatorial properties of the function, such as the existence of monochromatic rectangles.

Novelty

- Judge: `INCONCLUSIVE` over 15 arXiv hits

Top hits consulted: - [2311.03743v2] A general framework for the analytic Langlands correspondence - [2008.02998v3] Coherent sheaves on the stack of Langlands parameters - [2302.00039v1] Between Coherent and Constructible Local Langlands Correspondences - [1512.07347v1] Galois Self-Dual Constacyclic Codes - [1004.4962v1] Galois lines for normal elliptic space curves, II

Empirical Test

- exit code: `0`, elapsed: `0.04s`

```
n=5 has_rect=True order=1
n=8 has_rect=False order=0
n=11 has_rect=False order=0
n=14 has_rect=False order=0
RESULT: INCONCLUSIVE <reason>
```

Judge reasoning

Test results show partial alignment with conjecture but lack sufficient coverage to confirm or refute the general case. | next: Test larger n values and edge cases where rectangle size is minimal